

Multiple Choice (10 marks)

1. Which among the following is the least strong security encryption standard?
 - (a) WEP
 - (b) WPA
 - (c) WPA 2
 - (d) WPA 3
2. What was the first buffer overflow attack?
 - (a) Love Bug
 - (b) SQL Slammer
 - (c) Morris Worm
 - (d) Code Red
3. Buffer-overflow may remain as a bug in apps if _____ are not done fully.
 - (a) boundary hacks
 - (b) memory checks
 - (c) boundary checks
 - (d) buffer checks
4. What tool can be used to perform SNMP enumeration?
 - (a) DNSlookup
 - (b) Whois
 - (c) Nslookup
 - (d) IP Network Browser
5. A 3-way, set-associative cache is
 - (a) one in which each main memory word can be stored at any of 3 cache locations
 - (b) effective only if 3 or fewer processes are running alternately on the processor
 - (c) possible only with write-back
 - (d) faster to access than a direct-mapped cache

True / False (10 marks)

Answer True or False

6. True or False: A MAC can be secure even if it produces a short output, provided the underlying algorithm is strong.
7. True or False: Operator precedence in a programming language can be effectively defined using a context-free grammar.
8. True or False: EXE restarts the STP solver and retries the query until a solution is found, regardless of the timeout.
9. True or False: Buffer overflows occur in C and C++ due to the absence of string boundary checks in predefined functions.
10. True or False: Public key encryption is generally faster than symmetric key cryptography due to its simpler algorithmic processes.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

11. Write a function to calculate the sum of the positive numbers of a given list of numbers using lambda function.
12. Write a function to check if a nested list is a subset of another nested list.

End of Paper